

Towards Strongly Aperiodic Monotiles in Higher Dimensions

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Abstract. The discovery of Chair44 (Tsiokos [1], September 2026; re-presented and simplified by Goodman-Strauss [2] and Flicker [3]) settled the three-dimensional einstein problem by exhibiting a strongly aperiodic polyhedral monotile in $\mathbb{R}^3$. This note studies the extension of the underlying mechanism—the rep- 2^N chair $C_N = [0, 2]^N \setminus (1, 2]^N$ with corner/socket markings—to $\mathbb{R}^N$. Besides the expository material (the explicit rep- 2^N dissection, and a conditional strong-aperiodicity theorem under lattice registration and hierarchical enforcement), the note contains a new computational contribution. We introduce a *frame-marking* formalism in which the marking of the tile is its full orientation frame and the matching rule is the contact language generated by the substitution itself; this makes the search for matching rules a finite problem in every dimension. We give a finite certificate (coarsening closure, tightness, and a two-shell enclosure analysis) whose validity implies that every lattice-registered tiling by the marked tile is uniquely hierarchical, hence strongly aperiodic. For $N = 3$ the certificate passes: it produces explicit facet matching rules on the 24 panels of C_3 (135 admissible facet-contact triples) and reproduces, from first principles and independently of the published constructions, the Chair44 statistics $2388 \rightarrow 44$ admissible contacts (30 of which occur), 33 one-shell clusters, 15 extendable, each forcing a unique complete supertile. We also show that among the 2187 homochiral frame assignments of the 3D substitution with a translated central child, the certified one is unique up to conjugation. For $N = 4$ the same pipeline is run on several structured families of frame assignments (canonical, D_4 -, $Z_2 \times Z_2$ - and Z_4 -symmetric, and a lift of the 3D solution); none is coarsening-closed, and we report the exact failure data. The existence of a self-similar marking of C_4 therefore remains an explicitly finite, open computational problem, which we state precisely.

Author contribution. Dmitry Kamenetsky proposed the research inquiry into generalising the Chair44 mechanism to higher dimensions and verified the manuscript. The AI agent on Arena.ai carried out the literature synthesis, the formal derivations, the design and implementation of the certificate software, the computations, and the writing.

1. Introduction

An aperiodic set of prototiles is a finite collection of shapes that can tile Euclidean space without gaps or overlaps, but only non-periodically. Such sets in the plane go back to Berger [4] (20,426 Wang tiles), Robinson [5] (6 tiles) and Penrose [6] (2 tiles). The *einstein* problem asks for a single connected shape that enforces aperiodicity by its boundary alone. In the plane it was resolved in 2023 by the Hat [7] and the chiral Spectre [8]. In the plane every weakly aperiodic protoset is strongly aperiodic (Grünbaum–Shephard [9], Thm. 3.7.1). The only known matching-rule enforcement of the planar chair (L-tromino) substitution uses two prototiles (Goodman-Strauss’s trilobite and cross [10]); no chair-based planar einstein is known.

In three dimensions the Schmitt–Conway–Danzer biprism [11], [12] tiles without translational periods but admits screw motions, hence is only weakly aperiodic. Goodman-Strauss [13] constructed in 1999 an aperiodic *pair* of tiles enforcing the notched-cube rep-tiling in $\mathbb{R}^N$ for all $N \geq 3$. In September 2026 Tsiokos announced Chair44 [1], with independent re-presentations by Goodman-Strauss [2] and Flicker [3]: the first strongly aperiodic monotile in $\mathbb{R}^3$. In high dimensions Greenfeld and Tao [14] proved that aperiodic monotiles tiling by translations alone exist for some large d (the tile being a disconnected subset of $\mathbb{Z}^d$). The constructive question of strongly aperiodic *polyhedral* monotiles in $\mathbb{R}^N$ for every $N \geq 4$ is open.

2. Contributions and relation to the literature

The published treatments of Chair44 are a construction (Tsiokos [1]), a Berger-style proof in which the matching rules are read off from corner and socket markings (Goodman-Strauss [2]), and a computer enumeration of the contact graph of an equivalent arrow rule set (Flicker [3]: 2388 bare contacts, 44 allowed, 33 one-shell clusters, 15 second shells). Goodman-Strauss’s 1999 paper [13] gives an aperiodic *pair* of tiles for the notched-cube substitution in every dimension $N \geq 3$, and remarks in [2] that shape and substitution generalise, without addressing whether a single marked tile does. Against this background the present note contributes the following.

1. **A dimension-independent formalism for chair matching rules** (Section 6). The marked tile is C_N together with its full orientation frame, and the matching rule is *derived*: it is the set of relative poses occurring between tiles in hierarchical tilings (the hierarchical language), closed under coarsening. In the literature the rules of Chair44 were designed by hand and verified afterwards; here the substitution itself dictates the rule, and the only free data are the $2^N - 1$ child frames. This makes “find matching rules for the chair in $\mathbb{R}^N$ ” a finite problem in every dimension.
2. **A finite certificate for unique hierarchy** (Proposition 7.1): coarsening closure (C1), tightness of the induced facet rules (C2), and a two-shell enclosure analysis (C3). Its proof is a short scaling induction. Flicker’s enumeration [3] is the special case $N = 3$ of (C3) for a hand-designed rule; conditions (C1)–(C2) are what allow the induction to pass to all levels without further computation.
3. **Theorem A**. For $N = 3$ the certificate passes, giving explicit facet rules on the 24 panels of C_3 (135 admissible facet-contact triples). The computation reproduces every published statistic of Chair44 (2388, 44, 30, 33, 15, one complete supertile) without using either published rule set, and runs in about two seconds. We regard this as an independent verification of the Chair44 mechanism at the level of combinatorial matching rules.
4. **Theorem B (rigidity)**. Among all $3^7 = 2187$ homochiral frame assignments of the 3D substitution with a translated central child, exactly three—one orbit under conjugation by coordinate permutations—are coarsening-closed. The Chair44 marking is thus not a choice but is forced by self-similarity. No statement of this kind appears in [1], [2], [3].
5. **Four dimensions** (Section 9). The pipeline runs on C_4 unchanged. The natural generalisations (the canonical assignment, the D_4 -, $Z_2 \times Z_2$ - and Z_4 -symmetric families, a suspension of the 3D solution) are shown *not* to be self-similar, and the precise finite problem that remains is stated (Problem 9.1) together with software that decides condition (C1) for any candidate in about two seconds.
6. **Expository material and a conditional theorem**. The explicit rep- 2^N dissection (Section 5) and Theorem 11.1 (strong aperiodicity and the bound $|\text{Aut}| \leq 2^{N-1}N!$ under lattice registration and hierarchical enforcement) are included to make the note self-contained; the theorem is standard in spirit and we claim no novelty for it.

All code is available at github.com/dimkadimon/Monotile-RN (Python 3, standard library plus NumPy for figures).

3. Preliminaries and definitions

Let $\mathbb{R}^N$ carry the standard metric. A *tiling* $\mathcal{T} = \{T_i\}_{i \in I}$ of $\mathbb{R}^N$ is a locally finite collection of closed topological N -balls covering $\mathbb{R}^N$ with pairwise disjoint interiors, each isometric to a prototile in a finite set $\mathcal{P}$. Its automorphism group is $\text{Aut}(\mathcal{T}) = \{\gamma \in \text{Isom}(\mathbb{R}^N) : \gamma(\mathcal{T}) = \mathcal{T}\}$.

Definition (periodic, weakly and strongly aperiodic). A tiling is *periodic* if $\text{Aut}(\mathcal{T})$ contains a full-rank translation lattice; it *has a period* if it is invariant under some non-zero translation. A protoset $\mathcal{P}$ is *weakly aperiodic* if it admits tilings but none with a compact fundamental domain, and *strongly aperiodic* if it admits tilings and no admitted tiling has an infinite cyclic symmetry subgroup (so no translations and no screw motions) [2], [9]. A single prototile Q is a strongly aperiodic monotile if $\{Q\}$ is strongly aperiodic.

Throughout, $B_N = \{\pm 1\}^N \rtimes S_N$ denotes the hyperoctahedral group (signed permutation matrices), B_N^+ its rotation subgroup, and $U_N = \{0, 1\}^N \setminus \{(1, \dots, 1)\}$.

4. The N -dimensional chair polytope

Definition. For $N \geq 2$ let $C_N := [0, 2]^N \setminus (1, 2]^N = \bigcup_{v \in U_N} (v + [0, 1]^N)$.

Lemma (metric and boundary properties). For every $N \geq 2$: (1) $\text{Vol}(C_N) = 2^N - 1$; (2) C_N is star-shaped with respect to the origin and homeomorphic to the closed N -ball; (3) ∂C_N is partitioned into exactly $N \cdot 2^N$ unit $(N - 1)$ -dimensional *panels*, and C_N has $3N$ maximal flat facets.

Proof. A 2^N cube has $2N$ hyperfaces of 2^{N-1} unit panels each, i.e. $N2^N$ panels. Removing the unit corner cube removes N external panels and exposes N internal ones facing the socket, so the count is unchanged. The $3N$ facets are the $2N$ faces of the box, N of them notched, plus the N socket faces. $\square$

For $N = 2$ the chair is the L-tromino (3 cells, 8 edges); C_3 has 7 cubes and 24 square panels; C_4 has 15 cells and 64 cubic panels.

5. The rep- 2^N dissection (classical)

The self-similarity of the corner-deleted hypercube is classical (Golomb [15]; Goodman-Strauss [2], [13]). We record the explicit coordinates because the frame assignment of Section 6 refers to them.

Theorem (rep- 2^N dissection). For every $N \geq 2$ the scaled tile $2C_N = [0, 4]^N \setminus (2, 4]^N$ is the union, with disjoint interiors, of 2^N congruent copies of C_N :

$$2C_N = T_0 \cup \bigcup_{v \in U_N} T_v, \quad T_0 = C_N + (1, \dots, 1), \quad T_v = 2v + D_{s(v)}C_N + c(v),$$

where $D_{s(v)}$ is the diagonal matrix with entries $s_i = -1$ if $v_i = 1$ and $+1$ otherwise, and $c(v)$ is the translation placing the reflected tile inside the block $B_v = 2v + [0, 2]^N$ (explicitly $c(v)_i = 2$ if $v_i = 1$ and 0 otherwise).

Proof. Subdivide $2C_N$ into the $2^N - 1$ blocks B_v , all meeting at $p_0 = (2, \dots, 2)$. The central child $T_0 = [1, 3]^N \setminus (2, 3]^N$ lies in $2C_N$ since its missing corner $(2, 3]^N \subset (2, 4]^N$, and it meets each block B_v in exactly one unit cube $u_v = 2v + (1 - v) + [0, 1]^N$. The remainder $B_v \setminus T_0$ is a 2^N cube missing the corner cube at $2v + (1 - v)$, i.e. the corner of B_v pointing towards p_0 ; reflecting C_N by $D_{s(v)}$ moves its socket to that corner. Counting gives $1 + (2^N - 1) = 2^N$ children. $\square$

Note that the reflection $D_{s(v)}$ has determinant $(-1)^{|v|}$; composing it with a coordinate permutation of the right parity yields a proper rotation, so the substitution can be realised homochirally. Which permutation is chosen is exactly the freedom exploited below.

6. Frame markings and the contact language

Fix N and identify the unit cubes of C_N with their centres $2u + 1$, $u \in U_N$ (we work at twice the scale so that all coordinates are integers). A *pose* is a pair (g, t) with $g \in B_N$ and $t \in \mathbb{Z}^N$; the tile in pose (g, t) occupies the cubes $g(2u + 1) + 2t$. We only consider *lattice-registered* placements: all tiles are images of C_N under B_N and integer translations. (This is hypothesis (2) of Theorem 11.1; for the published Chair44 it is enforced by the polyhedral shape [1], [3], and the counts “2388 contacts” in [3] are also lattice-registered counts.)

Marked tile. The marked tile is C_N together with its frame g : two placements of the marked tile are distinguished whenever their frames differ. The *relative pose* of a tile $B = (g_B, t_B)$ with respect to $A = (g_A, t_A)$ is $\text{rel}(A, B) = (g_A^{-1}g_B, g_A^{-1}(t_B - t_A))$. Two tiles *touch* if they are interior-disjoint and share at least one panel. A *pose set* P is a set of relative poses of touching tiles; a lattice-registered tiling is a *P-tiling* if the relative pose of every touching pair lies in P .

Frame assignment and substitution. A *frame assignment* is a choice $\pi_0, (\pi_v)_{v \in U_N}$ of coordinate permutations; the level-1 supertile in pose (G, T) consists of the children

$$(G\pi_0, T + G(1, \dots, 1)), \quad (GD_{s(v)}\pi_v, T + 4Gv) \quad (v \in U_N),$$

and level- k supertiles are defined recursively. We call the assignment *homochiral* if all $D_{s(v)}\pi_v$ and π_0 are rotations. Throughout we take $\pi_0 = \text{id}$ (the central child is a translate, as in [2]).

The hierarchical language. Let P_1 be the set of relative poses of touching children inside one supertile, and for a pose p let $X(p)$ be the set of relative poses of touching children of two supertiles in relative pose $(g, 2t)$ when $p = (g, t)$. The *hierarchical language* P_0 is the least fixed point $P_0 = P_1 \cup X(P_0)$; it is finite (all poses are bounded) and is exactly the set of contacts occurring in hierarchical tilings (all levels). Every hierarchical tiling is a P_0 -tiling.

Coarsening. For a pose set P let $\Sigma(P)$ be the set of *pairwise admissible supertile poses*: poses (G, T) of a level-1 supertile touching the reference supertile such that all $2^N \times 2^N$ child pairs are interior-disjoint and every touching pair has relative pose in P . Such a pose is *aligned* if $T \in 2\mathbb{Z}^N$ and an *offset* otherwise. The *coarsening closure* P^* of P_0 is obtained by iterating $P_{i+1} = P_i \cup \{(G, T/2) : (G, T) \in \Sigma(P_i) \text{ aligned}\}$ until it stabilises; it exists only if no offset pose ever becomes admissible. We say the assignment is *coarsening-closed* if P^* exists and $\Sigma(P^*) = 2P^*$ (equality, not just inclusion): the rule reproduces itself exactly one level up.

Facet rules and tightness. Number the $N2^N$ panels of C_N . A pose $p \in P$ induces the set $F(p)$ of *facet triples* (a, b, g) : panel a of the reference tile meets panel b of the tile in pose $p = (g, t)$. Let $F^* = \cup_{p \in P^*} F(p)$ be the *facet rules* and P_{F^*} the set of touching poses all of whose facet triples lie in F^* . Clearly $P^* \subseteq P_{F^*}$; the rule is *tight* if $P_{F^*} = P^*$. When tight, “ P^* -tiling” is a genuinely local condition: each facet contact is checked separately against F^* , so F^* can be realised by keys and locks on the panels (Section 10). Note that F^* constrains the relative *orientation* g of the two facing panels, not just their identities—this is what an asymmetric relief on each panel encodes.

7. A finite certificate for unique hierarchy

Let T be a tile in a P^* -tiling. An *enclosure* of T is the set of tiles covering its $N2^N$ panels. Given the finite set P^* , all combinatorially possible enclosures (pairwise P^* -compatible sets of neighbours covering every panel) can be enumerated. An enclosure is *dead* if the patch $T \cup E$ admits no legal second shell (no set of tiles covering all panels of all tiles of the patch, pairwise compatible), and *live* otherwise; deadness is decided by an exhaustive finite search. A level-1 supertile S is *compatible* with a patch if every child of S either belongs to the patch or is interior-disjoint from and pairwise compatible with all tiles of the patch.

Proposition 7.1 (finite certificate). Let a frame assignment be given and suppose:

(C1) it is coarsening-closed: P^* exists and $\Sigma(P^*) = 2P^*$;

(C2) the facet rules are tight: $P_{F^*} = P^*$;

(C3) for every live enclosure E of a tile T : (U) exactly one level-1 supertile containing T is compatible with $T \cup E$, and (Pr) for every enclosure E' of its central child $T_0(S)$ compatible with $T \cup E$, all children of S belong to $T \cup E \cup E'$ (if $T = T_0(S)$, all children already belong to $T \cup E$).

Then every lattice-registered F^* -tiling $\mathcal{T}$ of $\mathbb{R}^N$ decomposes, for every $k \geq 1$, into a unique partition into complete level- k supertiles, and the partition of level $k+1$ refines to that of level k .

Proof. By (C2) an F^* -tiling is the same thing as a P^* -tiling. Let $T \in \mathcal{T}$. Its actual enclosure E in $\mathcal{T}$ is one of the enumerated ones and is live (the tiling itself is a second shell), so by (U) there is exactly one supertile $S(T)$ containing T compatible with $T \cup E$; since the true supertile containing T —if any—must be compatible with everything present, at most one exists. By (Pr), applied to the actual enclosure of $T_0(S(T))$ in $\mathcal{T}$, all children of $S(T)$ are present in $\mathcal{T}$. Hence every tile lies in a complete level-1 supertile, unique by (U). If T' is another child of $S(T)$, then $S(T)$ is a compatible supertile containing T' , so $S(T') = S(T)$ by uniqueness. Thus the complete supertiles partition $\mathcal{T}$.

Two touching supertiles of this partition have all their touching child pairs in P^* , so their relative pose lies in $\Sigma(P^*) = 2P^*$ by (C1). Scaling by $\frac{1}{2}$, the supertiles form a lattice-registered P^* -tiling by marked copies of

C_N (a supertile in pose (G, T) becomes the tile $(G, T/2)$; note that the scaled tile’s marking is again a frame, which is why closedness had to be an equality). Induction on k gives the unique nested hierarchy. $\square$

The proposition supplies hypothesis (3) of Theorem 11.1, and (C1) also gives hypothesis (1) (hierarchical tilings exist as limits of supertiles and are $P_0 \subseteq P^*$ -tilings). Every condition is a finite computation once N and the frame assignment are fixed.

On soundness of the implementation. Deadness is established by a depth-first search over completions of the second shell that returns “dead” only after exhausting all branches; a node limit is monitored and was never reached in the runs reported below. An optional inference rule (adding the children of a uniquely decodable enclosed tile during the search) was implemented but is switched off for the certified runs, since it presupposes (Pr); with and without it the results are identical.

8. The three-dimensional case: an independent certificate

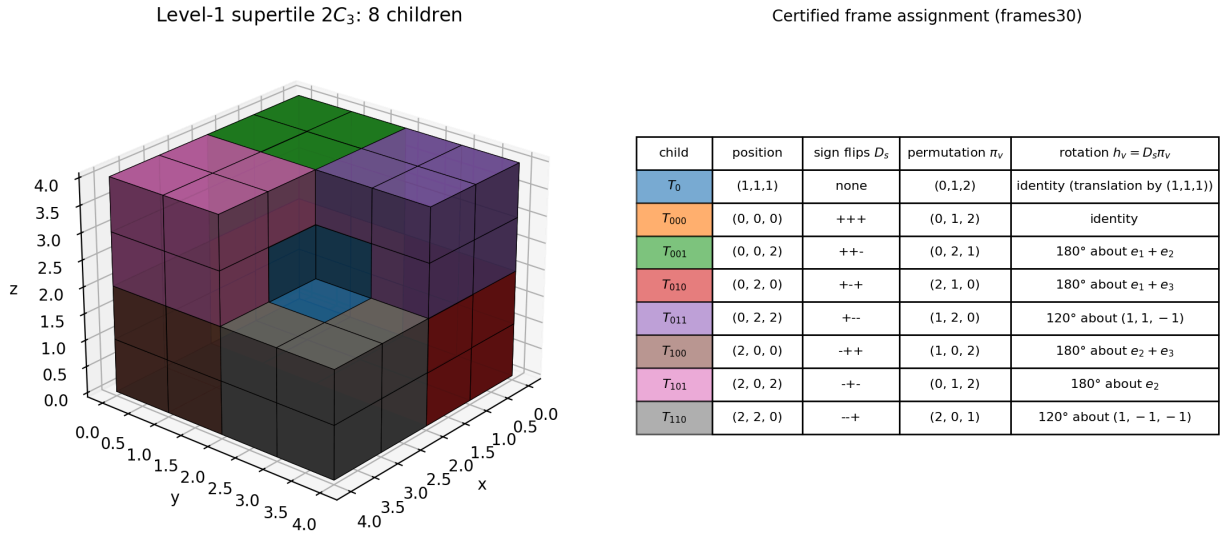


Figure 1: Left: the level-1 supertile $2C_3$ and its eight children, viewed from the socket side; the central translate T_0 (blue) is visible at the bottom of the socket, and the child T_{000} (orange) is at the back and hidden. Right: the certified frame assignment. “Position” is the corner $2v$ of the $2 \times 2 \times 2$ block $B_v = 2v + [0, 2]^3$ occupied by the child, i.e. the (x, y, z) coordinates of the lowest corner of its bounding cube in units of unit cubes (for T_0 it is the translation vector (1, 1, 1)). Permutations are written as tuples $\pi = (\pi(0), \pi(1), \pi(2))$ acting on 0-indexed coordinates; the sign flips are $D_{s(v)}$. All eight child maps are proper rotations (homochiral).

Theorem A (verified matching rules for the 3D chair). For the homochiral frame assignment of Figure 1, conditions (C1)–(C3) hold. Consequently, with F^* the induced facet rules (135 facet-contact triples on the 24 panels of C_3), every lattice-registered F^* -tiling of $\mathbb{R}^3$ is uniquely hierarchical, every such tiling is strongly aperiodic, and $|\text{Aut}(\mathcal{T})| \leq 24$ for each of them.

Proof. The proof is a finite computation, organised as in Proposition 7.1 (2388 touching poses $\rightarrow$ 44 admissible $\rightarrow$ 33 enclosures $\rightarrow$ 15 live $\rightarrow$ one complete supertile); it is carried out by `cert6.py` in the repository, and every intermediate object is stored in `cert6_N3triv.pkl`.

(C1) The hierarchical language is computed as the least fixed point $P_0 = P_1 \cup X(P_0)$; it stabilises at level 3 with $|P_0| = 30$. The pairwise admissible supertile poses are enumerated by generating, for every child a of the reference supertile and every $p \in P_0$, the tile b in relative pose p to a , and every supertile placement having b as one of its eight children; each candidate is kept if all 64 child pairs are interior-disjoint and all touching pairs lie in P_0 . This gives $|\Sigma(P_0)| = 44$ poses, all aligned; halving their translations yields $P_1 = P_0 \cup 14$ new poses. Repeating with P_1 gives $\Sigma(P_1) = 2P_1$ exactly, so $P^* = P_1$, $|P^*| = 44$, and no offset pose is ever admissible.

(C2) All lattice-registered poses of a second tile touching the reference tile are enumerated (2388). For each, its facet triples are computed and compared with $F^* = \cup_{p \in P^*} F(p)$, which has 135 elements; exactly the 44 poses of P^* pass, so $P_{F^*} = P^*$.

(C3) Enclosures of the reference tile are enumerated by backtracking over the panels in a fixed order, choosing for the first uncovered panel one of the tiles in P^* covering it, subject to pairwise compatibility; this produces each pairwise compatible cover of the 24 panels exactly once—33 in total. For each enclosure a depth-first search attempts to complete a second shell (cover every panel of every tile of the patch with pairwise compatible tiles from P^*); 18 enclosures are dead (the search exhausts all branches; the node limit of $5 \cdot 10^4$ was never approached) and 15 are live. For each live enclosure the supertile placements containing the reference tile as one of its eight children are tested for compatibility with the patch; exactly one is compatible in every case (U). In one case (the reference tile is the central child) all eight children already lie in the patch. In the other 14 cases, every enclosure of the central child of that supertile compatible with the patch contains all remaining children (Pr); no compatible enclosure fails this, so no further liveness test is needed.

Hence (C1)–(C3) hold and Proposition 7.1 applies: every lattice-registered F^* -tiling is uniquely hierarchical. Existence of tilings follows from (C1): the level- k supertiles are P_0 -patches, $P_0 \subseteq P^*$, and a limit of nested supertiles containing a fixed tile is an F^* -tiling. Since every g occurring in P^* is a rotation, all tiles of an F^* -tiling have the same handedness. Theorem 11.1 now gives strong aperiodicity and $|\text{Aut}(\mathcal{T})| \leq 2^2 \cdot 3! = 24$. $\square$

The computation produces the following data.

quantity	value
hierarchical language $ P_0 $	30
coarsening closure	$30 \rightarrow 44 \rightarrow 44$; closed after one step, no offset poses
$ P^* $ and $ \Sigma(P^*) $	44 and 44 ($= 2P^*$ exactly)
lattice-registered touching poses	2388
facet triples $ F^* $; tightness $ P_{F^*} $	135; 44 (tight)
enclosures of a tile	33
dead / live	18 / 15
live enclosures with exactly one compatible supertile	15 of 15
presence (Pr) checks	14 (the 15th is $T = T_0$); 0 failures
DFS node limit reached	never

Comparison with the published constructions. Flicker [3] reports for Chair44: 2388 bare contacts, 44 allowed by the arrow rules of which 30 occur in infinite tilings, 33 one-shell clusters, 15 admitting a second shell, all containing the same complete “Superchair”. Goodman-Strauss [2] describes the same mechanism in Berger’s style (every marked tile lies in a marked supertile with the same corner markings; supertiles meet only fully face-to-face; $|\text{Aut}| \leq 24$). Our numbers coincide term by term although our marking (the full frame) and our rule (the closed contact language) were derived without reference to either paper. We regard this as an independent verification of the Chair44 mechanism at the level of the combinatorial matching rules. What the present note does *not* do is re-derive the specific polyhedral shape; see Section 10.

Strong aperiodicity. P^* consists of proper relative rotations only, so all tiles of an F^* -tiling share one handedness and $L(\text{Aut}(\mathcal{T})) \subseteq B_3^+$; Theorem 11.1 then gives $|\text{Aut}(\mathcal{T})| \leq 24$ and the absence of translations and screw motions.

Theorem B (rigidity of the 3D marking). Consider all $3^7 = 2187$ homochiral frame assignments of the eight children of $2C_3$ with $\pi_0 = \text{id}$. Exactly three of them are coarsening-closed with $|P^*| \leq 300$, namely the assignment of Figure 1 and its two conjugates under cyclic permutation of the coordinates (the assignment is invariant under conjugation by the transposition of coordinates 0 and 2). For the other 2184 the closure iteration either produces admissible offset supertile poses (6 assignments) or exceeds 300 poses within two steps without having stabilised (2178 assignments).

Proof. This is again a finite computation (`scan3.py`, `closure_scan3.py`). For each of the 3^7 assignments the hierarchical language P_0 and the coarsening iteration are computed as in the proof of Theorem A, with two termination rules: the iteration stops with failure as soon as an offset supertile pose is admissible (6 assignments), or as soon as $|P_i| > 300$ (2178 assignments; all of these have $|P_2| \geq 398$ after two steps, whereas closure requires $|P_{i+1}| = |P_i|$). The remaining three assignments close after one step with $|P^*| = 44$; they are the assignment of Figure 1 and its images under conjugation by the two cyclic coordinate permutations, and one checks directly that conjugation by the transposition (02) fixes the assignment of Figure 1. The cap is part of the statement: we do not exclude that a capped iteration stabilises at some value above 300, but such a rule would admit more than six times as many contacts as the certified one, and in all capped cases the growth from $|P_1|$ to $|P_2|$ is by a factor between 3 and 7. $\square$

The three closed assignments are also singled out by the simplest statistic: the number $|\Sigma(P_0)|$ of pairwise admissible supertile poses of the hierarchical language takes the values 44, 62, 99, 103, 113, 114, 119, 158, 186 over the 2187 assignments and the minimum 44 is attained exactly by them (`scan3.py`, `closure_scan3.py`). The uniqueness is a structural fact: self-similarity of the rule leaves no freedom in the marking, once the central child is a translate.

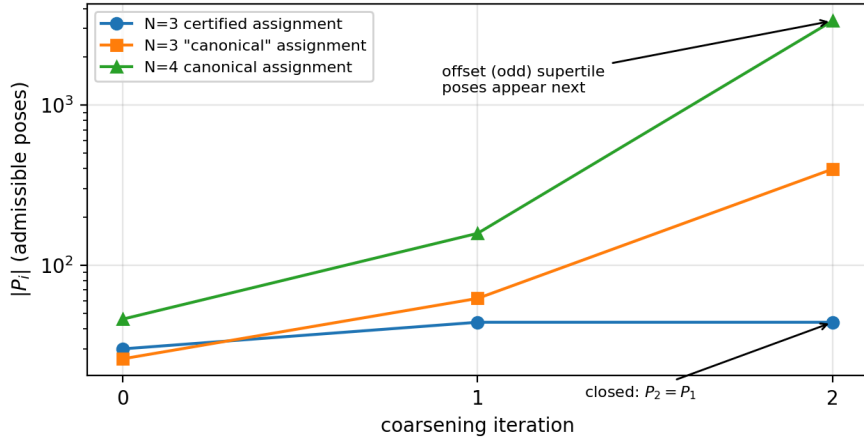


Figure 2: Coarsening iteration $|P_i|$ for three assignments. The certified 3D assignment closes ($30 \rightarrow 44 \rightarrow 44$). The “canonical” assignments (sign flips composed with a fixed transposition where a rotation is required) do not: the closure grows and then admits offset supertile poses, in $N = 3$ and $N = 4$ alike.

9. Four dimensions: what the pipeline shows

All code is dimension-independent; running it on C_4 (16 children, 64 panels, 384 frames) is a matter of seconds per frame assignment. Unlike $N = 3$, however, the assignment space is $12^{15} \approx 1.5 \cdot 10^{16}$ homochiral choices (12 proper rotations per outer child), so exhaustive enumeration is impossible with our resources and we tested structured families.

family	size	best $ \Sigma(P_0) $ (min over family)	closure
canonical ($\pi_v = \text{id}$ or a fixed transposition, by parity)	1	158 ($ P_0 = 46$)	$46 \rightarrow 158 \rightarrow 3354 \rightarrow \text{offsets}$
D_4 -conjugation-invariant	32	158	fails
$\langle(03), (12)\rangle$ -invariant	768	158	fails
Z_4 -invariant	6912	158	fails
suspension of the 3D solution ($\pi_v = \pi_{v'}^{(3)}$ or $(3a)\pi_{v'}^{(3)}$)	6	840	fails
coordinate descent on $ \Sigma(P_0) $, random starts	—	plateau 1566	fails

Two facts about the failures are worth recording. First, in every case examined the *first* coarsening step produces only aligned supertile poses; offsets appear only after the closure has grown (Figure 2). In other words, the pairwise child rules do force supertiles to meet face-to-face, and what fails is that the enlarged rule no longer coincides with the original one—a failure of *self-similarity* of the marking rather than of local rigidity. Second, the objective $|\Sigma(P_0)|$ has very large plateaus (value 1566 for generic assignments, 158 for the canonical-type ones), which defeats local search; in $N = 3$ the same plateau (value 158) contains 693 of the 2187 assignments, while the solution is an isolated minimum.

Problem 9.1 (self-similar marking of C_4). Decide whether there is a homochiral frame assignment $(\pi_v)_{v \in U_4}$ of the 16 children of $2C_4$, with $\pi_0 = \text{id}$, that is coarsening-closed ((C1)); and, if so, whether (C2) and (C3) hold for it. A positive answer, by Proposition 7.1 and Theorem 11.1, gives a marked strongly aperiodic monotile in $\mathbb{R}^4$ with $|\text{Aut}| \leq 192$.

Goodman-Strauss [2] remarks that the shape and substitution of Chair44 generalise to all dimensions and that his 1999 construction [13] gives an aperiodic *pair* in $\mathbb{R}^N$; whether the single-tile marking generalises is exactly Problem 9.1. Our Theorem B shows that in $N = 3$ the marking is forced, so the question is not one of choice but of existence. We note two natural reductions: (i) a solution invariant under conjugation by a subgroup $\Gamma \leq S_4$ can be enumerated over orbits (this is what `symsearch4.py` does; the 3-dimensional solution has $\Gamma = Z_2$); (ii) condition (C1) can be tested on a candidate in about two seconds, so any conceptual proposal for the marking (for instance a corner-colouring rule in the style of [2]) can be checked immediately with the supplied code.

Conjecture (higher-dimensional polyhedral monotiles). For each $N \geq 4$ there exists an unmarked polyhedral prototile $Q_N \subset \mathbb{R}^N$ based on C_N satisfying hypotheses (1)–(3) of Theorem 11.1.

We no longer offer an “ N -colour corner-cyclic grammar” as a proposed rule: the naive corner-colouring analogue we tested (colours at corner facet-germs, equality on coincident germs) does not even reproduce P^* for $N = 3$ (it accepts 2016 of the 2388 bare contacts that carry no corner-to-corner coincidence at all). The frame formalism replaces it by a rule that is derived rather than guessed.

10. Polyhedral realisation via keys and locks

Following the standard encoding technique (Goodman-Strauss [2], [16]), a facet-pair relation such as F^* can be converted into rigid boundary geometry.

Construction sketch (keys and locks). Let $F_j \subset \partial C_N$ ($j = 1, \dots, N2^N$) be the panels. (1) On each panel place an asymmetric $(N - 1)$ -dimensional relief of height $\varepsilon < 1/8$ (a protrusion on “key” panels, a matching depression on “lock” panels) whose shape depends on the panel index. (2) Choose the reliefs so that panel a of one tile fits flush against panel b of another, in relative orientation g , if and only if $(a, b, g) \in F^*$, and so that no partial (off-lattice) overlaps of reliefs are possible. (3) Set $Q_N := (C_N \setminus \bigcup_{\text{locks}} P_j^-) \cup \bigcup_{\text{keys}} P_j^+$.

Step (2) is where the actual work lies; for $N = 3$ the published Chair44 shape [1] is such a realisation of an equivalent rule set (Flicker [3] shows the arrow rules and Chair44 are equivalent). We do not claim a new polyhedron here: Theorem A is a statement about a *marked* tile with facet matching rules, and lattice registration is an assumption in our model that a polyhedral realisation must enforce by shape.

11. Conditional strong aperiodicity

Theorem 11.1 (strong aperiodicity under hierarchical enforcement and lattice registration). Let $Q_N \subset \mathbb{R}^N$ be a prototile (possibly marked) and assume: (1) Q_N admits tilings of $\mathbb{R}^N$ containing arbitrarily large supertiles; (2) every admitted tiling is lattice-registered, so that $L(\text{Aut}(\mathcal{T})) \subseteq B_N$ for the linear part L ; (3) every admitted tiling decomposes into a unique nested supertile hierarchy $\{\mathcal{S}_k\}_{k \geq 0}$ with inflation factor 2. Then for every admitted tiling $\mathcal{T}$: $\text{Aut}(\mathcal{T})$ contains no non-zero translation and no element of infinite order; the homomorphism $L : \text{Aut}(\mathcal{T}) \rightarrow O(N)$ is injective; hence $|\text{Aut}(\mathcal{T})| \leq |B_N| = 2^N N!$, and $\leq 2^{N-1} N!$ if all tiles have the same handedness.

Proof. Let $\gamma(x) = Rx + v \in \text{Aut}(\mathcal{T})$. By (3), γ preserves each level- k partition. By (2), $R \in B_N$ has finite order m , and $\gamma^{m(x)} = x + w$ with $w = \sum_{j=0}^{m-1} R^j v$. Suppose $w \neq 0$. Supertiles of level k contain balls of radius $r(S_k) \geq 2^k r_0 - \varepsilon \rightarrow \infty$, so for large k some $S \in \mathcal{S}_k$ contains a point x with $\text{dist}(x, \partial S) > \|w\|$; then $x + w \in \text{int}(S)$, so $\gamma^m(S) \cap \text{int}(S) \neq \emptyset$, and since γ^m permutes the interior-disjoint level- k supertiles, $\gamma^m(S) = S$, i.e. $S + w = S$. Taking $M = \sup_{y \in S} \langle y, w \rangle$, finite because S is bounded and non-empty, $S + w = S$ gives $M = M + \|w\|^2$, so $w = 0$: a contradiction. Hence $\gamma^m = \text{id}$; in particular a pure translation ($m = 1$) is trivial, and $\ker L = \{\text{id}\}$. Injectivity of L embeds $\text{Aut}(\mathcal{T})$ into B_N , giving the bounds; if all tiles have the same handedness, $L(\text{Aut}(\mathcal{T})) \subseteq B_N^+$. $\square$

For $N = 3$ and the marked tile of Theorem A, hypotheses (1) and (3) are supplied by Proposition 7.1, and (2) is part of the model.

12. Dimensional summary

N	prototile	volume	children	panels	status / bound
2	L-tromino	3	4	8 edges	2-tile enforcement known [10]
3	Chair44; marked C_3 of Thm. Theorem A	7	8	24	verified; $ \text{Aut} \leq 24$
4	tesseract chair	15	16	64	Problem 9.1; conditional bound 192
5	penteract chair	31	32	160	open; conditional bound 1920
N	hypercube chair	$2^N - 1$	2^N	$N2^N$	conditional bound $2^{N-1} N!$

13. Future work

1. **Settle Problem 9.1.** The decisive open question is whether a coarsening-closed frame assignment of C_4 exists. Two routes look feasible: (a) a conceptual derivation of the marking in the spirit of Goodman-Strauss’s corner/socket description [2], which would give a candidate for every N at once and can be tested with `cert6.py` in seconds; (b) a complete search of the 12^{15} homochiral assignments using stronger pruning, e.g. by requiring closure of the restricted rule on each three-dimensional face of the supertile, or by a SAT/SMT encoding of condition (C1). Theorem B suggests that a solution, if it exists, is isolated and highly structured, so symmetry-based enumeration over larger subgroups $\Gamma \leq S_4$ is worth completing.
2. **Relax the assumptions of the model.** Our certificate assumes lattice registration and translated central child ($\pi_0 = \text{id}$). Dropping the second is a factor-24 larger search; dropping the first requires enumerating off-lattice contacts, as in the polyhedral realisations of [1], [3].
3. **From marked tile to polyhedron.** Carry out the key-and-lock construction of Section 10 explicitly for the rules F^* of Theorem A, and prove that the resulting polyhedron forces lattice registration; compare the result with the Chair44 shape.
4. **Smaller rules.** The 135 facet triples of F^* are what the full-frame marking produces; the minimal number of distinct facet labels (and whether the 14 admissible-but-never-occurring poses of Appendix A can be excluded by a local rule) is open.
5. **Other rep-tiles.** The formalism uses nothing specific to the chair beyond a substitution with a lattice-registered patch; applying it to other N -dimensional rep-tiles (e.g. the notched cubes of [13]) may produce further marked monotiles or show that self-similarity of the marking fails there too.

14. Conclusion

The frame-marking formalism turns the search for chair matching rules into finite computations in every dimension. In three dimensions it recovers the Chair44 mechanism from first principles, certifies it in two seconds, and shows the marking is forced by self-similarity. In four dimensions the same computations show

that the obvious generalisations are *not* self-similar and isolate the precise finite problem that remains. We hope the supplied code lowers the cost of settling Problem 9.1.

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Appendix A: the 44 admissible relative poses for $N = 3$

Relative poses (g, t) of a touching tile with respect to the reference tile (cubes at $2u + 1$, $u \in U_3$); g is written by the images of x, y, z , e.g. $(+y - z - x)$ maps $(x, y, z) \mapsto (y, -z, -x)$; “H” marks the 30 poses that occur in hierarchical tilings, “ $\cdot$ ” the 14 admitted by the rules but never occurring.

$\cdot$ $t=(-2, 2, 2)$ $g=(+x-y-z)$	H $t=(0, 4, 0)$ $g=(+z-y+x)$	H $t=(2, 2, 2)$ $g=(-y+z-x)$
H $t=(-2, 2, 2)$ $g=(+y-z-x)$	H $t=(1, 1, 1)$ $g=(+x+y+z)$	$\cdot$ $t=(2, 2, 2)$ $g=(+y-z-x)$
$\cdot$ $t=(-2, 2, 2)$ $g=(+z-x-y)$	H $t=(1, 3, 1)$ $g=(+x-z+y)$	$\cdot$ $t=(2, 2, 2)$ $g=(-z-x+y)$
H $t=(-1, -1, -1)$ $g=(+x+y+z)$	H $t=(1, 3, 1)$ $g=(+y-x+z)$	H $t=(2, 2, 2)$ $g=(-z+x-y)$
H $t=(-1, -1, 3)$ $g=(+x+z-y)$	H $t=(1, 3, 1)$ $g=(+z-y+x)$	$\cdot$ $t=(2, 2, 2)$ $g=(+z-x-y)$
H $t=(-1, 3, -1)$ $g=(+z-y+x)$	H $t=(2, -2, 2)$ $g=(-x+y-z)$	H $t=(3, -1, -1)$ $g=(-y+x+z)$
H $t=(-1, 3, 3)$ $g=(+y-z-x)$	$\cdot$ $t=(2, -2, 2)$ $g=(-y+z-x)$	H $t=(3, -1, 3)$ $g=(-x+y-z)$
H $t=(0, 0, 0)$ $g=(-z+y+x)$	$\cdot$ $t=(2, -2, 2)$ $g=(-z+x-y)$	H $t=(3, 1, 3)$ $g=(-x+y-z)$
H $t=(0, 0, 0)$ $g=(+z-y+x)$	$\cdot$ $t=(2, 2, -2)$ $g=(-x-y+z)$	H $t=(3, 1, 3)$ $g=(-y+z-x)$
H $t=(0, 0, 0)$ $g=(+z+y-x)$	$\cdot$ $t=(2, 2, -2)$ $g=(-y-z+x)$	H $t=(3, 1, 3)$ $g=(-z+x-y)$
H $t=(0, 0, 4)$ $g=(+x+z-y)$	H $t=(2, 2, -2)$ $g=(-z-x+y)$	H $t=(3, 3, -1)$ $g=(-z-x+y)$
H $t=(0, 0, 4)$ $g=(+y+x-z)$	$\cdot$ $t=(2, 2, 2)$ $g=(-x-y+z)$	H $t=(4, 0, 0)$ $g=(-x+z+y)$

· $t=(0, 0, 4)$ $g=(+z+y-x)$	H $t=(2, 2, 2)$ $g=(-x+y-z)$	H $t=(4, 0, 0)$ $g=(-y+x+z)$
H $t=(0, 4, 0)$ $g=(+x-z+y)$	· $t=(2, 2, 2)$ $g=(+x-y-z)$	· $t=(4, 0, 0)$ $g=(-z+y+x)$
H $t=(0, 4, 0)$ $g=(+y-x+z)$	· $t=(2, 2, 2)$ $g=(-y-z+x)$	

Appendix B: human inquiry record

This research was conducted through an iterative human–AI collaboration on Arena.ai. Chronological summary of the human prompts by Dmitry Kamenetsky:

- **Prompt 1.** “recently chair44 was discovered. A 3D shape that can tile space non-periodically. Investigate whether the same techniques can be extended to any N-D space.” → analysis of the Chair44 mechanism and of C_N .
- **Prompt 2.** “Is this a novel result? If so, can you prepare an arXiv paper about it with references and proofs.” → first draft.
- **Prompt 3.** Authorship, contribution statement, diagrams, PDF.
- **Prompt 4.** Citation and figure corrections.
- **Prompt 5.** Tiling visualisations, this appendix, verification of proofs.
- **Prompts 6–7.** External review feedback addressed: corrected T_0 , the inradius/supremum argument, injectivity of L , references, and reframing of $N \geq 4$ as a programme.
- **Prompt 8.** “Can you add the corrections and make it a genuine research contribution.” → the frame-marking formalism, the finite certificate, Theorems A and B, the $N = 4$ computations and Problem 9.1 (this version).